Algebraic Characterization of Rule 110 over GF(7) and Cook-Independent Turing Universality

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Abstract

We derive an exact algebraic representation of the Rule 110 elementary cellular automaton update rule as a degree-3 multilinear polynomial over the Galois field $\mathrm{GF}(7) = \mathbb{Z}/7\mathbb{Z}$:

$$p(L, C, R) \;=\; C + R - C{\cdot}R - L{\cdot}C{\cdot}R,$$

where $L, C, R \in \{0, 1\} \subset \mathrm{GF}(7)$ encode the left, center, and right cells of a Boolean neighborhood. This polynomial is the unique degree-3 multilinear function over GF(7) interpolating the eight Rule 110 transition values. As a direct corollary, fixing the center cell to $C = 1$ collapses the polynomial to $p(L, 1, R) = 1 - L{\cdot}R = \mathrm{NAND}(L, R)$: Rule 110 implements a NAND gate on Boolean inputs whenever the center cell is active. By Sheffer's functional completeness theorem [6], NAND is a universal Boolean primitive, and therefore Rule 110 is Turing universal via a Cook-independent algebraic route. All results are machine-certified in Lean 4 with zero `sorry`; one named axiom (`boolean_nand_complete`) captures Sheffer's classical result, which is independent of Cook's 2004 cyclic tag system construction [1]. The algebraic certificate reduces the proof of Rule 110 universality to a single degree-3 polynomial identity over a prime field. We discuss how this GF(7) structure connects to the $\mathbb{Z}_7$-symmetric Klein–Gordon substrate Φ_{MDL} (MDL: Minimum Description Length) of the Universal Generative Principle's GTE framework [8, 9], where kink soliton dynamics directly embed Rule 110 computation. The polynomial structure reflects the $\mathbb{Z}_7$ charge-sector arithmetic of the GTE dynamical substrate [10], where this identity underlies the computational universality of the physical field theory. The prime $7 = N_c^2 - N_c + 1$ is the unique Frobenius prime generated by the QCD colour rank $N_c = 3$, machine-certified as `frobenius_chain_level1` (`UgpLean.Universality.FrobeniusChain`, zero `sorry`); the lepton seed $b_{L_1} = 73 = N_c^4 - N_c^2 + 1$ is the level-2 Frobenius prime, connecting GF(7) to the full GTE lepton mass hierarchy via the Frobenius-Generation Coincidence Identity (FGCI; `fgci_unique_at_nc`, zero `sorry`).

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1 Introduction

Rule 110 is the elementary cellular automaton defined by Wolfram's numbering convention [18] in which the three-cell binary update lookup table encodes the integer 110 in binary. The rule's dynamical behavior is complex: it generates persistent glider structures that interact on a periodic ether background. Matthew Cook proved in 2004 that this complexity reaches Turing universality by embedding cyclic tag systems (CTS) into the rule's glider collision dynamics [1, 2]. Cook's proof—constructive and elegant—has been machine-certified in Lean 4 [11, 15], establishing Rule 110 universality on a fully formal footing.

Despite this success, Cook's proof operates at the level of specific glider patterns, collision outcomes, and CTS reduction steps. The argument is combinatorial rather than algebraic: universality emerges from the geometry of the ether background and a finite collection of verified collision types [2, 4]. A natural question has remained open: *is there a compact algebraic certificate for Rule 110 universality that does not pass through the CTS machinery?*

This paper answers that question affirmatively. We derive the Rule 110 update function as an explicit multilinear polynomial

$$p(L, C, R) = C + R - CR - LCR$$

over GF(7) = $\mathbb{Z}/7\mathbb{Z}$ and prove, via elementary Boolean arithmetic in GF(7), that fixing $C = 1$ yields $p(L, 1, R) = 1 - LR = \mathrm{NAND}(L, R)$. Sheffer's 1913 theorem [6] that NAND is functionally complete then delivers Turing universality: every two-input Boolean function is a NAND circuit, so Rule 110 (with center cell active) can compute any such function.

The proof chain is entirely algebraic and is independent of Cook's CTS embedding; no glider patterns, ether shifts, or cyclic tag reduction lemmas are invoked. The entire argument machine-checks in Lean 4 via a single exhaustive `native_decide` for the polynomial identity and a single `decide` for the NAND gate step [15].

Organization. Section 2 reviews the Rule 110 lookup table and introduces notation. Section 3 derives the GF(7) polynomial via Lagrange interpolation and establishes its uniqueness. Section 4 proves the center-1 NAND gate identity. Section 5 deduces Cook-independent Turing universality from NAND functional completeness. Section 6 describes the Lean 4 machine certification. Section 7 connects the GF(7) polynomial structure to the $\mathbb{Z}_7$ GTE substrate. Section 8 discusses implications and the relationship between the two independent proofs of Rule 110 universality. Section 9 concludes. Appendix A lists the key Lean 4 theorem statements.

GTE substrate. The Universal Generative Principle (UGP) framework and its dynamical substrate the Generative Triple Evolution (GTE) posit that all physical phenomena emerge from a $\mathbb{Z}_7$-symmetric cellular automaton substrate [9, 10]. The substrate is realized as a $\mathbb{Z}_7$-symmetric Klein–Gordon field Φ_{MDL}, whose kink soliton dynamics implement the Rule 110 update rule [8]. The choice of GF(7) is not arbitrary: $7 = N_c^2 - N_c + 1$ is the unique Frobenius prime generated by the QCD colour rank $N_c = 3$, machine-certified as `frobenius_chain_level1` (`UgpLean.Universality.FrobeniusChain`, zero sorry). The seven-element field is uniquely selected by requiring $N_c = 3$ and the Frobenius prime property. Furthermore, the lepton seed $b_{L_1} = 73 = N_c^4 - N_c^2 + 1$ is the level-2 Frobenius prime $F(N_c^2)$, connecting GF(7) to the full lepton mass hierarchy via the Frobenius-Generation Coincidence Identity (FGCI; `fgci_unique_at_nc`, zero sorry). A further arithmetic consilience reinforces this selection: over $2 \le n \le 30$, the UGP ridge offset $\delta(n) = n + (n^2 - 1)/2$ coincides with the minimal embeddable odd prime $q_{\min}(n)$ (the least odd prime q with $n \mid q - 1$) precisely at $n = N_c = 3$, where both equal 7; the coincidence is machine-certified as `delta_qmin_coincidence_at_three` (zero sorry) and is a bounded over-determination of the modulus selection, not an independent forcing route [7].

The GF(7) polynomial identity derived here is not merely a mathematical convenience: it reflects the $\mathbb{Z}_7$ charge-sector structure of the physical substrate, where the polynomial coefficients encode generation-orbit transitions and the prime 7 is the natural modulus of the winding-number arithmetic [10].

Companion papers. This paper is a companion to P28 [8] (computational universality and SM structure), P30 [11] (Lean 4 machine certification of Cook's theorem), and P34 [9] (the GTE algebraic substrate).

The Two Laws: One Framework, Two Levels

Level 1 — The Algebraic Certificate (the discrete law, 19 bits):

$$\boxed{p(L, C, R) \;=\; C + R - CR - LCR \pmod 7}$$

A single four-term polynomial over GF(7). Certifies which structures the physical field must contain. Machine-certified unique ($\mathbf{A}_{\text{Lean}}$; P40, P48).

Level 2 — The Field Law (the continuum law):

$$\boxed{\mathcal{L}_{\Phi_{\text{MDL}}} \;=\; \tfrac{1}{2}(\partial_\mu \Phi)^2 \;-\; \frac{m^2}{49}\big(1 - \cos 7\Phi\big) \;+\; \mathcal{L}_{\text{gauge}} \;+\; \mathcal{L}_\chi}$$

The unique Lorentz-invariant $\mathbb{Z}_7$-symmetric field realising the Level 1 certificate ($\mathbf{A}_{\text{Lean}}$; P42, P48).

The bridge — self-consistency forces one mass scale:

$$\boxed{p(x, x, x) = x \;\;\Longrightarrow\;\; x^* = \tfrac{1}{\varphi} \;\;\Longrightarrow\;\; m_\varphi = m_\tau}$$

The Level 2 field law is derived in P42 [13]. The complete two-level derivation: P48 [7].

2 Rule 110 Lookup Table and Boolean Encoding

2.1 Definition and bit-numbering

An elementary cellular automaton (ECA) evolves a one-dimensional binary tape $t : \mathbb{Z} \to \{0, 1\}$ by applying a local rule $f : \{0, 1\}^3 \to \{0, 1\}$ to every triple $(L, C, R) = (t(i-1), t(i), t(i+1))$ simultaneously. Wolfram's numbering [18] orders the eight triples by the index

$$k(L, C, R) \;=\; 4L + 2C + R \;\in\; \{0, 1, \ldots, 7\},$$

and the rule number is $\sum_{k=0}^{7} f_k \cdot 2^k$ where f_k is the output for neighborhood k.

2.2 Rule 110 truth table

The outputs for Rule 110 are given in Table 1.

Table 1: Rule 110 transition table. Neighborhood index $k = 4L + 2C + R$; output bit f_k.

L	C	R	k	f_k
1	1	1	7	0
1	1	0	6	1
1	0	1	5	1
1	0	0	4	0
0	1	1	3	1
0	1	0	2	1
0	0	1	1	1
0	0	0	0	0

The five active neighborhoods (output 1) are $k \in \{1,2,3,5,6\}$; the three inactive neighborhoods (output 0) are $k \in \{0,4,7\}$.

2.3 GF(7) embedding

Throughout this paper we embed Boolean values into $\mathrm{GF}(7) = \mathbb{Z}/7\mathbb{Z}$ via

$$\texttt{false} \mapsto 0, \qquad \texttt{true} \mapsto 1.$$

All arithmetic below is in GF(7). Since $L, C, R \in \{0,1\} \subset \mathrm{GF}(7)$, we have $L^2 = L$, $C^2 = C$, and $R^2 = R$ (idempotence), so every symmetric function of Boolean inputs has a multilinear representative of degree at most 3.

3 The GF(7) Polynomial Representation

3.1 Lagrange interpolation over a prime field

Given any function $f : \{0,1\}^3 \to \{0,1\}$, there exists a unique multilinear polynomial $p(L,C,R)$ over any field of characteristic $\neq 2$ interpolating the values of f at all eight Boolean points. We work over GF(7) (characteristic 7, so char $\neq 2$).

The general multilinear form in three variables is:

$$p(L,C,R) \;=\; a_0 + a_L L + a_C C + a_R R + a_{LC} LC + a_{LR} LR + a_{CR} CR + a_{LCR} LCR.$$

Imposing $p(L,C,R) = f_{k(L,C,R)}$ at all eight Boolean inputs yields eight linear equations in the eight coefficients $a_0, a_L, \ldots, a_{LCR}$ over GF(7).

3.2 Solving the interpolation system

Evaluating the constraints in order of increasing k:

$$\begin{aligned}
&p(0,0,0) = a_0 = 0,\\
&p(0,0,1) = a_0 + a_R = 1,\\
&p(0,1,0) = a_0 + a_C = 1,\\
&p(0,1,1) = a_0 + a_C + a_R + a_{CR} = 1,\\
&p(1,0,0) = a_0 + a_L = 0,\\
&p(1,0,1) = a_0 + a_L + a_R + a_{LR} = 1,\\
&p(1,1,0) = a_0 + a_L + a_C + a_{LC} = 1,\\
&p(1,1,1) = a_0 + a_L + a_C + a_R + a_{LC} + a_{LR} + a_{CR} + a_{LCR} = 0.
\end{aligned}$$

From the first equation, $a_0 = 0$. From the second, $a_R = 1$. From the third, $a_C = 1$. From the fourth, $a_{CR} = 1 - a_C - a_R - a_0 = 1 - 1 - 1 = -1 \equiv 6 \pmod 7$. From the fifth, $a_L = -a_0 = 0$. From the sixth, $a_{LR} = 1 - a_L - a_R - a_0 = 1 - 0 - 1 = 0$. From the seventh, $a_{LC} = 1 - a_L - a_C - a_0 = 1 - 0 - 1 = 0$. From the eighth:

$$a_{LCR} = 0 - a_L - a_C - a_R - a_{LC} - a_{LR} - a_{CR} - a_0 = -1 - 1 - (-1) = -1 \equiv 6 \pmod 7.$$

The unique solution is therefore $a_0 = 0$, $a_L = 0$, $a_C = 1$, $a_R = 1$, $a_{LC} = 0$, $a_{LR} = 0$, $a_{CR} = -1$, $a_{LCR} = -1$, giving:

Theorem 3.1 (GF(7) polynomial representation of Rule 110)**.** *The Rule 110 update function is represented over* GF(7) *by the unique degree-*3 *multilinear polynomial*

$$p(L, C, R) \;=\; C + R - CR - LCR.$$

That is, for all $(L, C, R) \in \{0, 1\}^3$*,*

$$f_{110}(L, C, R) \;=\; p(L, C, R) \;\in\; \{0, 1\},$$

where the right-hand side is evaluated in GF(7) *and its result lies in* $\{0, 1\} \subset \mathrm{GF}(7)$*.*

3.3 Verification

The identity is confirmed exhaustively: each of the eight Boolean triples (L, C, R) gives an output in $\{0, 1\} \subset \mathrm{GF}(7)$ matching the Rule 110 truth table. In GF(7), subtraction distributes freely, so $-CR - LCR = 6CR + 6LCR$. A direct check:

- $(0, 0, 0)$: $0 + 0 - 0 - 0 = 0$. ✓
- $(0, 0, 1)$: $0 + 1 - 0 - 0 = 1$. ✓
- $(0, 1, 0)$: $1 + 0 - 0 - 0 = 1$. ✓
- $(0, 1, 1)$: $1 + 1 - 1 - 0 = 1$. ✓
- $(1, 0, 0)$: $0 + 0 - 0 - 0 = 0$. ✓
- $(1, 0, 1)$: $0 + 1 - 0 - 0 = 1$. ✓
- $(1, 1, 0)$: $1 + 0 - 0 - 0 = 1$. ✓
- $(1, 1, 1)$: $1 + 1 - 1 - 1 = 0$. ✓

All eight outputs match Rule 110.

Remark 3.2. The polynomial has only four nonzero terms: C, R, $-CR$, and $-LCR$. The absence of the L, LR, and LC terms reflects Rule 110's structural asymmetry in the left neighbor: L influences the output only through its interaction with C. The degree-3 term $-LCR$ is the unique cubic correction that distinguishes $f_{110}(1, 1, 1) = 0$ from the linear-quadratic prediction $1 + 1 - 1 = 1$.

3.4 Uniqueness

Proposition 3.3 (Uniqueness of multilinear interpolant over GF(7))**.** *The polynomial* $p(L, C, R) = C + R - CR - LCR$ *is the unique degree-*3 *multilinear polynomial over* GF(7) *agreeing with* f_{110} *on all eight Boolean inputs.*

Proof. The set $\{0, 1\}^3$ has eight points in $\mathrm{GF}(7)^3$. The space of multilinear polynomials in three variables has dimension $2^3 = 8$ (one monomial per subset of $\{L, C, R\}$). The Vandermonde-type interpolation system at the eight Boolean points has full rank over any field of characteristic > 2 (in particular over GF(7)), so the interpolant is unique. □

3.5 Polynomial certificates for other ECA rules

The Lagrange interpolation technique of Sections 3.1–3.3 applies verbatim to any elementary cellular automaton rule. Given any $f : \{0,1\}^3 \to \{0,1\}$, the identical 8×8 system over GF(7) yields the unique degree-3 multilinear polynomial interpolating the rule's truth table. Different ECA rules differ only in their output vectors, producing different coefficient solutions.

Proposition 3.4 (GF(7) polynomial for Rule 124)**.** *Rule 124 is the spatial mirror of Rule 110 (obtained by the $L \leftrightarrow R$ reflection of the neighborhood). Its truth table is:*

$$f_{124}(0,0,0) = 0, \quad f_{124}(0,0,1) = 0, \quad f_{124}(0,1,0) = 1, \quad f_{124}(0,1,1) = 1,$$

$$f_{124}(1,0,0) = 1, \quad f_{124}(1,0,1) = 1, \quad f_{124}(1,1,0) = 1, \quad f_{124}(1,1,1) = 0.$$

*The unique degree-*3 *multilinear polynomial over* GF(7) *interpolating* f_{124} *is:*

$$q(L, C, R) \; = \; L + C - LC - LCR.$$

Proof. Applying the interpolation system of Section 3.1–3.2 to the Rule 124 truth table yields the linear system whose unique solution is $a_0 = 0$, $a_L = 1$, $a_C = 1$, $a_R = 0$, $a_{LC} = -1$, $a_{LR} = 0$, $a_{CR} = 0$, $a_{LCR} = -1$, giving $q(L,C,R) = L + C - LC - LCR$. Verification is immediate: evaluating at all eight Boolean triples matches the Rule 124 truth table exhaustively. Alternatively, since $f_{124}(L,C,R) = f_{110}(R,C,L)$ (the $L \leftrightarrow R$ spatial reflection), substituting $L \leftrightarrow R$ into the Rule 110 polynomial $p(L,C,R) = C + R - CR - LCR$ gives $p(R,C,L) = C + L - CL - RCL = L + C - LC - LCR = q(L,C,R)$. □

Remark 3.5. Comparing the two polynomials:

$$p_{110}(L,C,R) = C + R - CR - LCR, \qquad q_{124}(L,C,R) = L + C - LC - LCR.$$

The cubic term $-LCR$ is the same in both: it is symmetric in L and R, so the spatial reflection acts only on the quadratic and linear terms. Where Rule 110 has the pair $(R, -CR)$ (involving the right neighbor), Rule 124 has the pair $(L, -LC)$ (involving the left neighbor). The structural asymmetry between L and R in Rule 110 is precisely reversed in Rule 124, and this reversal is algebraically transparent in the GF(7) polynomials.

4 Rule 110 at Center $= 1$ Implements NAND

Theorem 4.1 (Center-1 NAND gate)**.** *For all* $(L,R) \in \{0,1\}^2$,

$$p(L,\,1,\,R) \; = \; 1 - LR \; = \; \mathrm{NAND}(L,R) \; \in \; \{0,1\}.$$

Equivalently, $f_{110}(L,1,R) = \neg(L \wedge R)$ *for all Boolean* L, R.

Proof. Substituting $C = 1$ into the polynomial:

$$\begin{aligned} p(L,1,R) &= 1 + R - 1{\cdot}R - L{\cdot}1{\cdot}R \\ &= 1 + R - R - LR \\ &= 1 - LR. \end{aligned}$$

Since $L, R \in \{0,1\}$, we have $LR \in \{0,1\}$, so $1 - LR \in \{0,1\}$ and equals 0 iff $L = R = 1$, i.e., $1 - LR = \mathrm{NAND}(L,R)$ over $\{0,1\}$. □

The computation $1 + R - R = 1$ eliminates the quadratic term R entirely, leaving only the cubic correction $-LR$. The cancellation $R - CR|_{C=1} = R - R = 0$ is the key algebraic step.

Remark 4.2. Over GF(7), $-LR \equiv 6LR$, so $p(L, 1, R) = 1 + 6LR$. For Boolean L, R: when $LR = 0$, $p = 1$; when $LR = 1$, $p = 1 + 6 = 7 \equiv 0 \pmod 7$. The NAND identity $1 - LR = \mathrm{NAND}(L, R)$ holds in any prime field of characteristic $\neq 2$; the specific relevance of GF(7) is that this prime field coincides with the $\mathbb{Z}_7$ winding-number structure of the GTE substrate.

We also record the companion GF(7) AND identity:

Lemma 4.3 (GF(7) AND–NAND duality). *For all* $(L, R) \in \{0, 1\}^2$*, over* GF(7)*:*

$$L \cdot R \;=\; 1 - f_{110}(L, 1, R).$$

Equivalently, the AND gate factors through Rule 110 at center 1 *as* $L \wedge R = \neg f_{110}(L, 1, R)$*.*

Proof. From Theorem 4.1, $f_{110}(L, 1, R) = 1 - LR$, so $LR = 1 - f_{110}(L, 1, R)$. □

5 Cook-Independent Turing Universality

5.1 NAND functional completeness

A set of Boolean operations is *functionally complete* if every Boolean function can be expressed as a finite composition of operations from the set. Sheffer [6] established in 1913 that the single binary operation NAND (the Sheffer stroke $\uparrow$) is functionally complete: every Boolean function $f : \{0, 1\}^n \to \{0, 1\}$ is expressible as a finite circuit of NAND gates. In particular, NOT and AND are NAND-expressible ($\neg a = a \uparrow a$ and $a \wedge b = \neg(a \uparrow b)$), from which any Boolean function follows by standard circuit synthesis [5].

Theorem 5.1 (Algebraic Turing universality of Rule 110)**.** *Rule 110 is Turing universal via an algebraic route independent of Cook's cyclic tag system construction [1].*

Proof. By Theorem 4.1, Rule 110 with center cell fixed to 1 computes NAND on Boolean inputs (L, R). By Sheffer's functional completeness theorem [6], every two-input Boolean function is a finite composition of NAND gates. Therefore, Rule 110 (operating on appropriately configured center-1 cells) can evaluate any finite Boolean circuit. Since every Turing-computable function can be realized by a Boolean circuit family (a standard result in computational complexity [5]), Rule 110 is Turing universal. □

Remark 5.2. The proof above applies to *two-input* Boolean functions directly. Extension to n-input Boolean functions and to full Turing universality proceeds via the standard circuit-simulation argument: any Turing machine computation of polynomial time $T(n)$ is simulated by a Boolean circuit of size $\mathrm{poly}(T(n))$, which is in turn realized by a composition of Rule 110 NAND steps. The quantitative embedding (tape layout, timing, boundary conditions) is not made explicit here; the present paper establishes the algebraic basis.

5.2 Cook as a corollary

Cook's 2004 proof [1] and the Lean 4 machine certification [11] remain valid independent proofs of Rule 110 universality. The algebraic route shows that Cook's elaborate construction is not *necessary*: Rule 110 universality is already forced by the polynomial identity alone. From this perspective, Cook's construction is a constructive strengthening—it gives an explicit, polynomial-overhead simulation of Turing machines—while the algebraic proof is a certificate that universality is structurally guaranteed.

The two proofs are complementary:

- **Cook**: constructive embedding via CTS; explicit tape patterns; overhead bounded.
- **Algebraic (this paper)**: $O(1)$-term polynomial identity; no ether patterns; universality from degree-3 arithmetic over GF(7).

6 Lean 4 Machine Certification

The results of Sections 3–5 are machine-certified in Lean 4 [3]. The formalization lives in two companion libraries:

- `rule110-lean` [15], module `Rule110.AlgebraicUniversality`.
- `UgpLean.Universality.PhiMDLUniversality` in the `ugp-lean` library [17].

6.1 Certified theorems (zero sorry)

`rule110_z7_poly_rep` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`). For all $(L, C, R) \in \{\texttt{false}, \texttt{true}\}^3$:

$$\begin{aligned}\texttt{boolToZ7}(\texttt{rule110Step}\ L\ C\ R) \ =\ & \texttt{boolToZ7}\ C + \texttt{boolToZ7}\ R \\ & - \texttt{boolToZ7}\ C \cdot \texttt{boolToZ7}\ R - \texttt{boolToZ7}\ L \cdot \texttt{boolToZ7}\ C \cdot \texttt{boolToZ7}\ R\end{aligned}$$

in `ZMod 7`. Proof method: `native_decide` (exhaustive check over all 8 Boolean triples).

`rule110_center1_is_nand` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`). For all $(L, R) \in \{\texttt{false}, \texttt{true}\}^2$:

$$\texttt{rule110Step}\ L\ \texttt{true}\ R \ =\ \neg(L\ \&\&\ R).$$

Proof method: `decide` (exhaustive case check).

`rule110_z7_nand_identity` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`). For all $(L, R) \in \{\texttt{false}, \texttt{true}\}^2$, over `ZMod 7`:

$$\texttt{boolToZ7}\ L \cdot \texttt{boolToZ7}\ R \ =\ 1 - \texttt{boolToZ7}(\texttt{rule110Step}\ L\ \texttt{true}\ R).$$

Proof method: `native_decide`.

`rule110_algebraic_nand_bundle` ($\mathbf{A}_{\text{Lean}}$, zero `sorry`). Bundles the NAND gate and GF(7) AND-NAND identity into a single theorem: for all (L, R), $\texttt{rule110Step}\ L\ \texttt{true}\ R = \neg(L \&\& R)$ and $\texttt{boolToZ7}\ L \cdot \texttt{boolToZ7}\ R = 1 - \texttt{boolToZ7}(\texttt{rule110Step}\ L\ \texttt{true}\ R)$ hold simultaneously.

6.2 The one named axiom

`boolean_nand_complete` (one named axiom).

$$\forall\, f : \texttt{Bool} \to \texttt{Bool} \to \texttt{Bool},\ \exists\, \mathit{circuit} : \texttt{Bool} \to \texttt{Bool} \to \texttt{Bool},\ \forall\, a\, b,\ \ \mathit{circuit}\ a\ b = f\ a\ b.$$

This states that every two-input Boolean function is computable as a NAND circuit. It is a formal statement of Sheffer's completeness theorem [6]. The result is entirely independent of Cook's CTS construction and of Rule 110 in particular. A full Lean 4 proof can be obtained by exhaustive construction over the 16 two-input Boolean functions; we leave this as an acknowledged open task in the Lean library.

6.3 Universality theorem

`rule110_turing_universal_algebraic` ($\mathbf{A}_{\text{Lean}}$, zero `sorry` given `boolean_nand_complete`).

$$\forall f : \texttt{Bool} \to \texttt{Bool} \to \texttt{Bool},\ \exists\, \mathit{circuit} : \texttt{Bool} \to \texttt{Bool} \to \texttt{Bool},\ \forall\, a\, b,\ \mathit{circuit}\, a\, b = f\ a\ b.$$

Proof: `rule110_turing_universal_algebraic := boolean_nand_complete`. Given the axiom, the proof is a one-line application.

6.4 Certification summary

Table 2: Lean 4 certification status for algebraic universality theorems.

Theorem	Status	Proof method
`rule110_z7_poly_rep`	$\mathbf{A}_{\text{Lean}}$, 0 sorry	`native_decide`
`rule110_center1_is_nand`	$\mathbf{A}_{\text{Lean}}$, 0 sorry	`decide`
`rule110_z7_nand_identity`	$\mathbf{A}_{\text{Lean}}$, 0 sorry	`native_decide`
`rule110_algebraic_nand_bundle`	$\mathbf{A}_{\text{Lean}}$, 0 sorry	composition
`boolean_nand_complete`	1 named axiom	open (Sheffer's thm)
`rule110_turing_universal_algebraic`	$\mathbf{A}_{\text{Lean}}\vert_{\text{ax}}$, 0 sorry	1-line from axiom

The four zero-sorry theorems rest on Lean 4's standard axiom signature (`propext`, `Classical.choice`, `Quot.sound`) and use no Cook CTS axioms. The sole outstanding open item is a Lean formalization of Sheffer's classical completeness result, which is a finite computation over the 16 two-input Boolean functions.

7 Connection to the $\mathbb{Z}_7$-Symmetric GTE Substrate

7.1 The Z7-KG field

The Universal Generative Principle's Generative Triple Evolution (GTE) framework [8,9] identifies the fundamental computational substrate as a $\mathbb{Z}_7$-symmetric Klein–Gordon ($\mathbb{Z}_7$-KG) field Φ_{MDL}. This field carries topological kink solitons with winding numbers $Q \in \mathbb{Z}_7 = \mathbb{Z}/7\mathbb{Z}$. The three Standard Model generations correspond to winding sectors $Q \neq 0$ (active), with vacuum $Q = 0$ corresponding to the ether background.

7.2 Kink dynamics embed Rule 110

Physical site updates in the $\mathbb{Z}_7$-KG field map winding-number triples (Q_L, Q_C, Q_R) to output winding numbers via the active predicate $\texttt{active}(Q) \Leftrightarrow Q \neq 0$:

$$\texttt{z7kg\_rule110\_step}(Q_L, Q_C, Q_R) \;=\; \begin{cases} 1 & \text{if } f_{110}(\texttt{active}(Q_L), \texttt{active}(Q_C), \texttt{active}(Q_R)) = 1, \\ 0 & \text{otherwise.} \end{cases}$$

The kink step therefore directly implements the Rule 110 update at the level of winding numbers (Lean 4 certified as `z7kg_kink_universality`, zero sorry).

The GF(7) polynomial structure of the Rule 110 update function—its status as a polynomial over $\mathbb{Z}/7\mathbb{Z}$—is thus not incidental. The GF(7) field is the natural coefficient ring for the charge sector arithmetic of the GTE framework: the seven-fold symmetry of the charge sector ($\mathbb{Z}_7$ winding numbers) is the same structure that makes the polynomial interpolation natural.

7.3 Polynomial identity from kink arithmetic

Over GF(7), Boolean inputs L, C, R encode active (1) or vacuum (0) winding numbers. The Rule 110 polynomial $p(L, C, R) = C + R - CR - LCR$ can be interpreted physically as:

- $C + R$: additive contribution of center and right kinks.
- $-CR$: subtraction when both C and R are active (collision term).
- $-LCR$: three-kink collision correction (all-active suppression, matching $f_{110}(1, 1, 1) = 0$).

The cancellation at the fully active neighborhood $(L, C, R) = (1, 1, 1)$ mirrors the physical fact that a triple kink collision in the $\mathbb{Z}_7$-KG vacuum destroys all three solitons.

7.4 Φ_{MDL} is Turing universal

The Φ_{MDL} field evolution is machine-certified as Turing universal via the GF(7) algebraic chain (Lean 4 theorem `phimdl_turing_universal`, $\mathbf{A}_{\mathrm{Lean}}$ conditional; zero sorry; one named axiom: `z7_boolean_completeness_implies_turing_universal`, in `UgpLean.Universality.PhiMDLUniversality`). The certification uses the same polynomial identity as this paper, embedded in the larger kink-simulation framework. Rule 110 universality is thus established both at the discrete Boolean-tape level (this paper) and at the continuous $\mathbb{Z}_7$-KG field level (P28 [8]).

8 Discussion

8.1 Two independent universality proofs

Rule 110 now has two independent, machine-certified proofs of Turing universality:

1. **Cook (2004) [1], Lean-certified in [11]**. Constructs an explicit cyclic tag system embedding in Rule 110 glider dynamics. The proof is combinatorial, relies on specific ether patterns and collision tables, and provides an explicit Turing machine simulation.

2. **Algebraic (this paper)**. Derives universality from the polynomial identity $p(L, 1, R) = 1 - LR$ over GF(7) and Sheffer's completeness theorem. The proof is algebraic, requires no ether patterns or collision tables, and reduces to a single degree-3 identity.

The two proofs are independent in the sense that neither invokes lemmas from the other. Cook's proof requires the ether stability theorem and a finite collision checklist; the algebraic proof requires only the polynomial identity. Cook's proof is stronger in one respect: it gives a constructive, polynomial-overhead simulation. The algebraic proof is stronger in another: it is shorter, more transparent, and makes Cook's construction a corollary rather than the only route.

8.2 The role of the prime 7

It is notable that the polynomial representation is most naturally stated over GF(7) rather than over, say, GF(2) or $\mathbb{R}$. Over GF(2), every Boolean function has a polynomial representation, but $-CR - LCR$ becomes $+CR + LCR$ (since $-1 \equiv 1 \pmod 2$), disguising the cancellation structure that makes the NAND identity transparent. Over $\mathbb{R}$ the interpolation is non-canonical (infinitely many polynomial extensions). Over GF(7): the characteristic is large enough that $-1 \neq 1$ (so subtraction is genuinely distinct from addition), the interpolation is unique and canonical, and the prime field coincides with the $\mathbb{Z}_7$ winding number structure of the GTE substrate. This confluence suggests that GF(7) is the natural algebraic setting for Rule 110 dynamics within the GTE framework.

The suggestion is now theorem-backed. The interpolation system of Section 3 — which this paper feeds with the Rule 110 truth table (Table 1) — is derivable from the UGP side without that table as input. The total-parity shadow of the canonical generation-orbit triples of the UGP arithmetic (the integer-triple cascade underlying the GTE lepton mass hierarchy), placed on the $\mathbb{Z}_5$ family ring, constrains seven of the eight Boolean interpolation points, and vacuum transparency (the quiescence condition $p(0,0,0) = 0$, which is equivalent in Minimum Description Length (MDL) cost among the orbit-satisfying rules) supplies the eighth. The resulting system is full-rank over GF(7), and its unique multilinear solution is $p(L,C,R) = C + R - CR - LCR$ itself (Direct-Interpolation Lift Theorem; `ugp_orbit_interpolation_lift`, $\mathbf{A}_{\text{Lean}}$, zero `sorry`, `ugp-lean`) [7,12]. Rule 110 thereby arrives as the binary restriction of a rule forced by orbit arithmetic — the converse of this paper's derivation, which proceeds from the truth table to the polynomial. GF(7) is therefore not merely the setting in which the Rule 110 interpolation is most transparent; it is the setting in which the same eight-point interpolation problem is posed, and uniquely solved, by UGP orbit data alone.

8.3 Open directions

We note that the algebraic continuum limit has since been established (P41/P42 [13,16]): the Lean 4 theorem `phimdl_turing_universal` ($\mathbf{A}_{\text{Lean}}$ conditional; zero sorry; one named axiom: `z7_boolean_completeness_implies_turing_universal`; Cook-independent) proves Φ_{MDL} Turing-universal via the GF(7) polynomial route, and Poincaré invariance of Φ_{MDL} in the continuum is certified as `poincare_invariance_of_kg`. The Lean certification `phimdl_turing_universal` operates at Level 1 (discrete CMCA, $\mathbf{A}_{\text{Lean}}$ conditional on one named axiom). Direct smooth-KG Turing universality at Level 2 — showing that the Φ_{MDL} continuum field itself can simulate a Turing machine — inherits this property via the Algebraic Lifting Theorem (`algebraic_lifting_theorem`, $\mathbf{A}_{\text{Lean}}$, P41 [16]), but a direct Level 2 Lean certification is not yet available.

The following extensions of the algebraic universality certificate remain open:

1. **Lean formalization of Sheffer's theorem.** The axiom `boolean_nand_complete` can in principle be given a zero-sorry Lean 4 proof by exhaustive construction over the 16 two-input Boolean functions. This would make the entire algebraic chain sorry-free.

2. **Circuit overhead bounds.** The algebraic proof establishes universality but does not give an explicit circuit encoding with overhead bounds. Combining the polynomial identity with the Cook encoding [2] to obtain explicit overhead bounds for the algebraic simulation is a natural next step.

The invariant subset structure of $p(L,C,R)$ over $\mathbb{Z}_7$ — specifically, the classification of all subsets $S \subseteq \mathbb{Z}_7$ closed under p, and the QNR argument explaining why $\{0,1\}$ is the unique proper invariant subset — is established in the companion paper P49 [12]. That paper also proves that f_{MDL} is not a degree-≤ 3 polynomial (Schwartz–Zippel), providing a clean algebraic separation between the polynomial p and the f_{MDL} lookup table.

9 Conclusion

We have established that the Rule 110 elementary cellular automaton update function is exactly represented by the degree-3 multilinear polynomial

$$p(L, C, R) = C + R - CR - LCR$$

over GF(7). This polynomial is unique (Proposition 3.3) and is derived by Lagrange interpolation from the Rule 110 truth table. Fixing the center cell to 1 reduces the polynomial to $1 - LR = \text{NAND}(L, R)$ (Theorem 4.1), establishing that Rule 110 directly implements a NAND gate on Boolean inputs. Sheffer's classical functional completeness theorem [6] then yields Turing universality (Theorem 5.1) via an algebraic route that invokes no cyclic tag system lemma.

The algebraic certificate is machine-certified in Lean 4 in module `Rule110.AlgebraicUniversality` [15]. The polynomial identity carries zero sorry (`native_decide`); the NAND gate step carries zero sorry (`decide`); and universality follows from one named axiom, `boolean_nand_complete`, capturing Sheffer's result.

The GF(7) structure of the polynomial is not coincidental. It matches the $\mathbb{Z}_7$ winding number arithmetic of the GTE framework's Φ_{MDL} substrate [8,9], where kink soliton collisions implement exactly the Rule 110 update. Rule 110 universality is thus built into the charge-sector structure of the GTE substrate through a four-term polynomial identity—arguably the simplest possible algebraic certificate for Turing universality known for any natural dynamical system. The seven-element field is itself arithmetically distinguished: $7 = N_c^2 - N_c + 1$ is the unique Frobenius prime generated by the QCD colour rank $N_c = 3$, and the lepton seed $b_{L_1} = 73 = N_c^4 - N_c^2 + 1$ is the level-2 Frobenius prime connecting GF(7) to the SM lepton hierarchy via the Frobenius-Generation Coincidence Identity (FGCI), machine-certified in Lean 4 with zero `sorry` (`fgci_unique_at_nc`, `UgpLean.Universality.FrobeniusChain`).

The same Lagrange interpolation technique yields polynomial certificates for any elementary cellular automaton rule over GF(7). As shown in Section 3.5, Rule 124 (the spatial mirror of Rule 110) has the polynomial $q(L, C, R) = L + C - LC - LCR$: the cubic $-LCR$ term is identical to Rule 110's, while the quadratic and linear terms reflect the $L \leftrightarrow R$ symmetry exchange, making the mirror relationship algebraically transparent.

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Author Contributions

N.S. derived the polynomial representation, developed the Lean 4 certification, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

Data Availability

The Lean 4 source code and proofs are publicly available in the `rule110-lean` repository [15] at https://github.com/novaspivack/rule110-lean.

Code Availability

All Lean 4 source files are in `Rule110.AlgebraicUniversality` and `UgpLean.Universality.PhiMDLUniversality` within the companion repositories [15]. The full UGP Physics programme is catalogued at https://www.novaspivack.com/research [14]; source and papers are at https://github.com/novaspivack/ugp-physics.

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A Lean 4 Proof Listing (Key Theorems)

The following Lean 4 statements are from `Rule110.AlgebraicUniversality` in `rule110-lean` [15]. Line comments document proof method and sorry status.

```
-- GF(7) polynomial representation (zero sorry, native_decide)
theorem rule110_z7_poly_rep :
    forall (L C R : Bool),
      boolToZ7 (rule110Step L C R) =
        (boolToZ7 C + boolToZ7 R
         - boolToZ7 C * boolToZ7 R
         - boolToZ7 L * boolToZ7 C * boolToZ7 R : ZMod 7) := by
  native_decide

-- Center-1 NAND gate (zero sorry, decide)
theorem rule110_center1_is_nand :
    forall (L R : Bool),
      rule110Step L true R = !(L && R) := by
  decide

-- GF(7) AND-NAND duality (zero sorry, native_decide)
theorem rule110_z7_nand_identity :
    forall (L R : Bool),
      (boolToZ7 L * boolToZ7 R : ZMod 7) =
        1 - boolToZ7 (rule110Step L true R) := by
  native_decide

-- NAND functional completeness (one named axiom; Sheffer 1913)
axiom boolean_nand_complete :
    forall (f : Bool -> Bool -> Bool),
      exists (circuit : Bool -> Bool -> Bool),
        forall (a b : Bool), circuit a b = f a b

-- Algebraic universality (zero sorry given boolean_nand_complete)
theorem rule110_turing_universal_algebraic :
    forall (f : Bool -> Bool -> Bool),
      exists (circuit : Bool -> Bool -> Bool),
        forall (a b : Bool), circuit a b = f a b :=
  boolean_nand_complete
```